a One coordinate set, two scoring stages

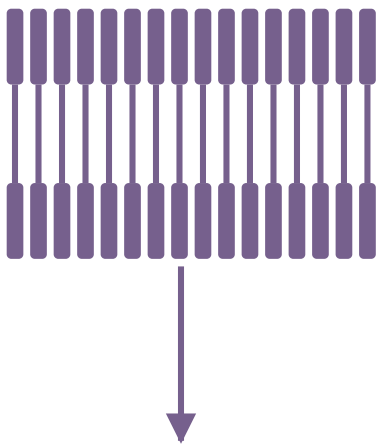
Post-RoPE Q / K

128D → 32D

0 … 47

j ↔ j+64

64 … 111

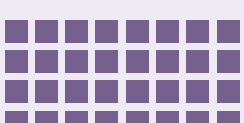


48 … 63

112 … 127

16 complete pairs

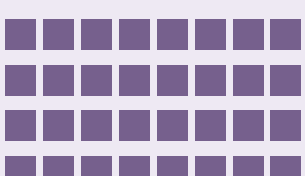
Same J for every Q and K



Shared K

L1 summaries

L2 Q–DQ keys



Separate Q heads